

Final Capstone Project

Top Spotify Songs in Venezuela

An End-to-End Data Analytics Project Using BigQuery, Python, and Power BI

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TEDA 2051 — Data Technology Program
Mountainland Technical College
June 2026

1. About the Data

The dataset used for this project is the **Top Spotify Songs in 73 Countries (Daily Updated)** dataset, sourced from Kaggle and originally created by scraping Spotify's official daily Top 50 charts. The full dataset is approximately 497 MB and includes daily chart rankings for 73 countries. For this project, the data was filtered to include only Venezuela (country code 'VE'), resulting in a working dataset of 28,263 rows.

Each row represents a single song's position on a single day's Top 50 chart in Venezuela. The dataset includes 25 columns covering identification fields (song name, artist, Spotify ID), chart performance metrics (daily rank, daily movement, weekly movement), and detailed audio features generated by Spotify's own analysis engine (danceability, energy, valence, tempo, acousticness, and more).

Key Terminology

- **Daily Rank:** The song's position (1-50) on a given day's chart.
- **Popularity:** A Spotify-generated score (0-100) reflecting overall listening activity.
- **Danceability:** A score (0-1) describing how suitable a track is for dancing.
- **Energy:** A score (0-1) representing intensity and activity in a track.

Limitations and Considerations

Because the dataset tracks the same Top 50 chart daily, songs that remain popular over long periods appear as multiple rows, which inflates simple row counts. Some early-2024 records contained missing song or artist names and were removed during cleaning. Additionally, the dataset reflects platform behavior (what people stream on Spotify) rather than full market behavior, including radio or physical sales.

2. Problem Statement

Streaming platforms generate massive volumes of chart and behavioral data, but raw rankings alone do not explain *why* certain songs and artists achieve sustained popularity. For stakeholders such as record labels, playlist curators, or marketing teams operating in the Venezuelan market, understanding which artists dominate the charts and what musical characteristics correlate with chart success can directly inform release strategy, marketing investment, and content curation decisions.

This project investigates the Venezuelan Spotify chart landscape to identify patterns in artist dominance, popularity trends over time, and the relationship between audio characteristics and chart performance.

3. Business / Research Questions

1. Who are the most frequently charted artists on Spotify Venezuela, and what makes them consistently popular?
2. How does average song popularity trend over time across the dataset's date range?

3. How do the top artists compare to one another in terms of average popularity?
4. How do audio features such as danceability and energy relate to one another among charting songs?
5. Which songs achieved the #1 position on the Venezuelan chart, and how popular are they on average?

4. Data Preparation

Data preparation followed a multi-stage cloud workflow designed to move from a large, unwieldy public dataset to a clean, analysis-ready table hosted in Google BigQuery.

Cleaning and Transformation Steps

- **Size reduction:** The original 497 MB, 73-country file exceeded the project's manageability guidelines. Using Python and Pandas, the dataset was filtered to Venezuela only ('VE'), reducing it to 28,263 rows and under 10 MB.
- **Cloud ingestion:** The filtered CSV was uploaded to Google BigQuery under the project capstone-guillermo, dataset spotify_data, creating the venezuela_top_songs table with auto-detected schema.
- **Missing value handling:** Rows with null song names or artist fields were identified and removed during the Colab EDA phase to ensure accurate aggregation.
- **Feature engineering:** New fields were created, including year and month (extracted from snapshot_date) and duration_sec (converted from milliseconds), to support time-based analysis.
- **Duplicate and consistency checks:** The dataset was checked for fully duplicated rows; none materially affected the analysis given the daily-snapshot structure of the data.
- **Re-publishing:** The cleaned DataFrame (df_clean) was written back to BigQuery as venezuela_clean, and a grouped summary table (top_songs) was created to support dashboard-level aggregations.

Assumptions Made

Analysis assumes that a song's repeated appearance across many daily snapshots is a meaningful proxy for sustained popularity rather than a data artifact. Collaborative tracks (e.g., "Bad Bunny, Feid") were treated as distinct artist-string entities rather than being split into individual contributors, consistent with how Spotify itself credits multi-artist tracks.

5. Analysis and Findings

Exploratory analysis was conducted in Google Colab using Python (Pandas, Matplotlib, Seaborn) connected directly to BigQuery. The analysis combined descriptive statistics, frequency analysis, and visualizations to answer the project's guiding questions.

Finding 1: Artist Dominance

Bad Bunny dramatically dominates the Venezuelan Spotify charts, appearing far more frequently in the Top 50 than any other artist, followed by KAROL G and Feid. Feid notably appears both as a solo artist and in frequent collaborations, making his overall presence on the chart even larger than a single-artist count would suggest. This indicates that a small handful of artists account for a disproportionate share of chart real estate in the Venezuelan market.

Finding 2: Popularity Trends Over Time

Average song popularity does not follow a steady or linear trend across the date range analyzed. Instead, it fluctuates substantially, with notable peaks in late 2023 and again in early 2025, interspersed with sharp declines. This suggests that chart popularity in Venezuela is influenced by periodic events (album releases, viral moments, or seasonal listening habits) rather than a gradual market shift.

Finding 3: Audio Feature Relationships

There is a visible positive relationship between danceability and energy among charting songs: tracks with higher danceability scores also tend to have higher energy scores. This pattern suggests that Venezuelan listeners favor upbeat, rhythmic, danceable music, which aligns with the genre mix dominated by reggaeton and Latin urban artists in the dataset.

Finding 4: #1 Songs and Popularity

Songs that reached the #1 position on the Venezuelan daily chart show a wide range of average popularity scores, from the high 80s and 90s (e.g., tracks by Bad Bunny and Feid) down to scores near zero for a handful of niche or short-lived #1 entries. This indicates that reaching #1 in Venezuela specifically does not always correlate with high global Spotify popularity, highlighting the value of analyzing a local market separately from global trends.

6. Dashboard Showcase

An interactive three-page dashboard was built in Power BI, connected directly to the cleaned BigQuery tables. The dashboard was designed using a Spotify-inspired color palette (green, black, and white) to reinforce the project's subject matter and improve visual clarity.

Page 1: Overview

This page provides at-a-glance KPIs (total songs, unique artists, average popularity) alongside a Top 10 artists bar chart. It answers the question of who dominates the Venezuelan chart and gives users an immediate sense of the dataset's scale before diving into deeper analysis.

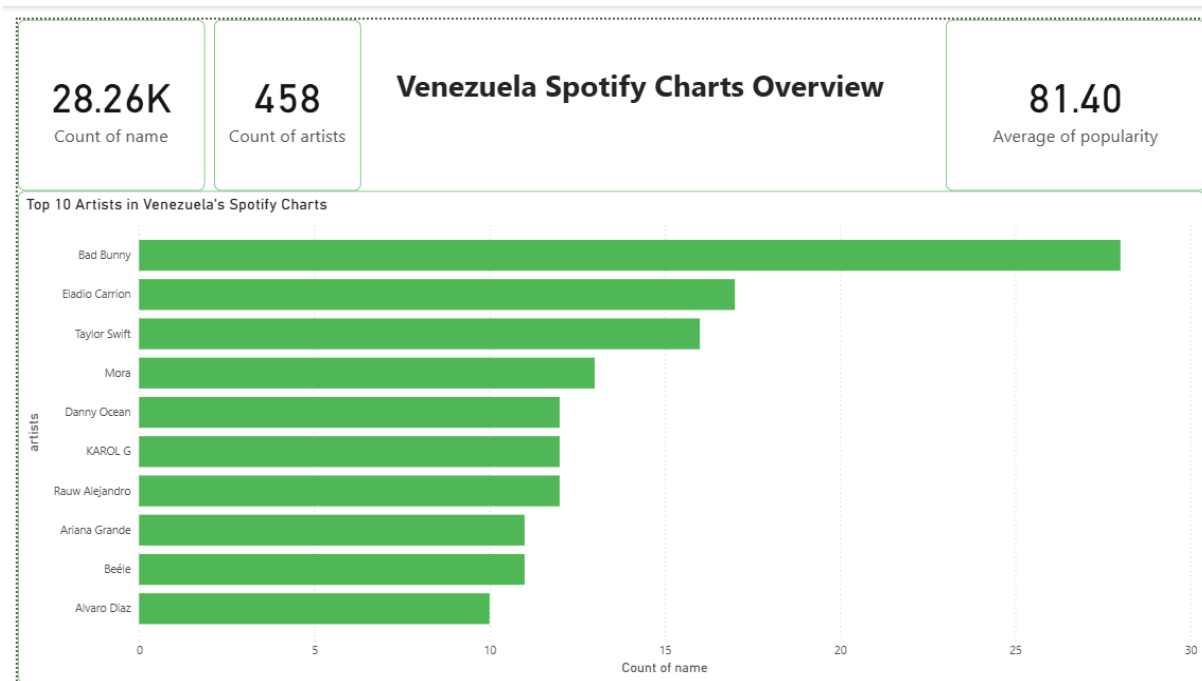


Figure 1. Overview page showing KPI cards and the Top 10 most frequently charted artists.

Page 2: Trends Over Time

This page compares the average popularity of the top charting artists over time using a multi-line chart, with an interactive date-range slider allowing users to zoom into specific periods. This visual directly supports Questions 2 and 3 by showing both overall popularity fluctuation and artist-to-artist comparison in a single interactive view.

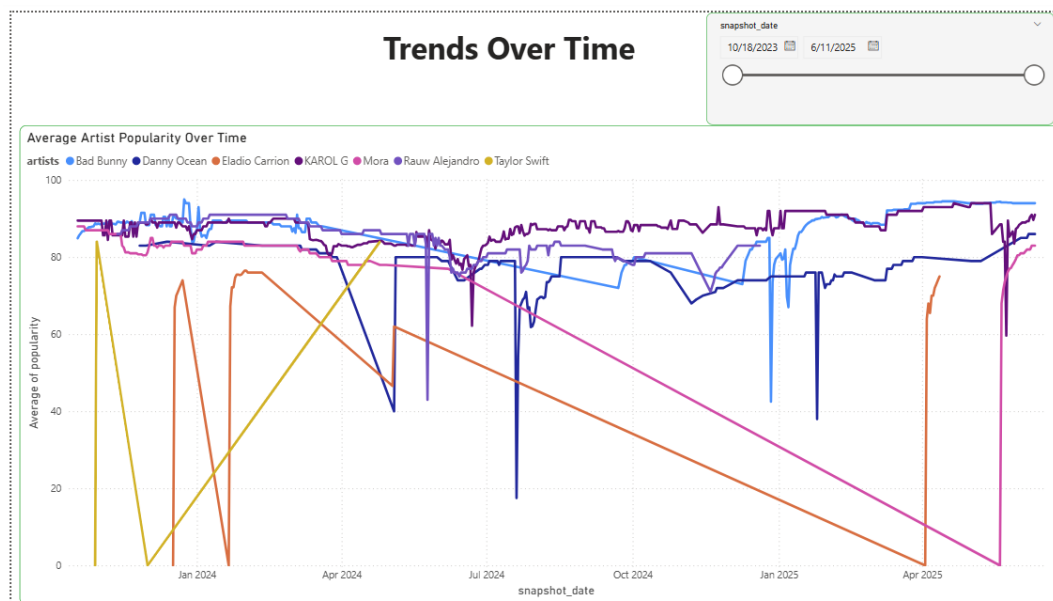


Figure 2. Trends page comparing average popularity across top artists, with an interactive date-range slicer.

Page 3: Audio Features Analysis

This page pairs a scatter plot of danceability versus energy with a sortable table of songs that reached the #1 position, including their artist and average popularity. Together, these visuals support Questions 4 and 5, allowing users to explore both the broad audio-feature pattern across the dataset and the specific songs responsible for the highest individual chart achievement.

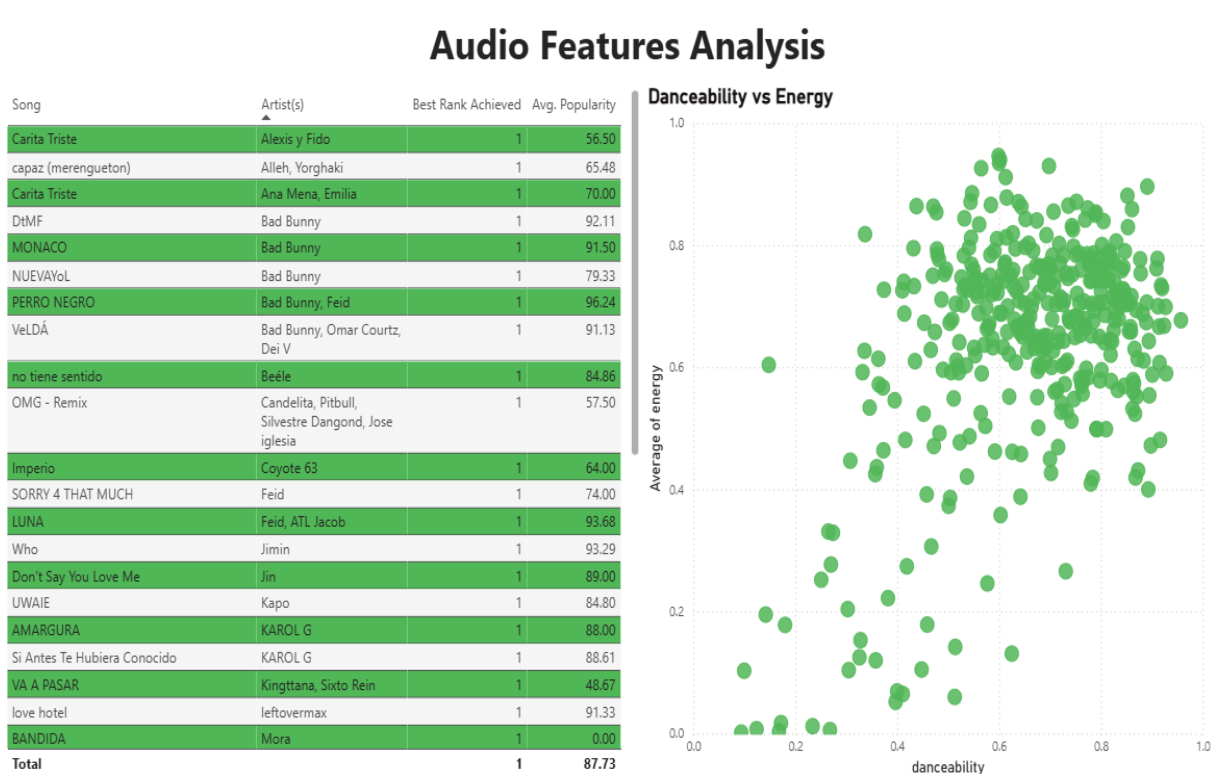


Figure 3. Audio features page showing the danceability-energy relationship and a table of #1 songs.

Dashboard Interactivity

The dashboard includes a date-range slicer on the Trends page that filters the line chart dynamically, allowing decision-makers to isolate specific periods (e.g., a single quarter or release window) without needing to rebuild the visual. This interactivity is central to the dashboard's purpose: supporting exploratory, self-service analysis rather than presenting a single static snapshot.

7. Conclusions and Recommendations

This project successfully traced a complete analytics pipeline — from a raw, oversized public dataset through cloud storage, Python-based exploratory analysis, and an interactive Power BI dashboard — to answer concrete questions about the Venezuelan Spotify chart landscape.

Major Findings

- A small group of artists, led by Bad Bunny, accounts for a disproportionate share of chart presence in Venezuela.

- Chart popularity fluctuates in bursts rather than trending steadily, suggesting event-driven rather than gradual shifts.
- Danceable, high-energy tracks are consistently overrepresented among charting songs.
- Reaching #1 locally does not guarantee high global popularity, underscoring the value of market-specific analysis.

Recommendations

- Marketing and playlist curation teams targeting the Venezuelan market should prioritize danceable, high-energy releases, particularly from established dominant artists, to maximize chart visibility.
- Release timing should be considered strategically, given the bursty nature of popularity trends observed in the data.
- Future work should incorporate genre classification and lyrical content analysis to better explain why specific songs outperform others beyond audio features alone.

Opportunities for Future Analysis

Future iterations of this project could expand to comparative analysis across multiple Latin American countries, incorporate predictive modeling to forecast chart movement, or integrate external data sources such as social media engagement to better explain the drivers behind the popularity spikes identified in this analysis.